

CE 604: Theory of Plasticity

Course Code	CE 604
Course Name	Theory of Plasticity
L-T-P-C	2-1-0-3
Prerequisites	Strength of Materials and Structures, Structural Analysis, Theory of elasticity, advanced solid mechanics
Intended for	P.G. (M.Tech Structure, Design, etc., M.Tech by Research, and Ph.D.)
Approval	52nd BoA (Nov 2023)

Course Contents

- **Introduction:** Fundamental principles of plasticity, Basic laws of plasticity, Index notations. (4 Lectures)
- **Criteria of yielding:** Maximum stress theory, Maximum strain theory, maximum shear theory, maximum strain energy theory, Distortion energy theory, Mohr's theory of yielding, Yielding surfaces. (6 Lectures)
- **Plastic stress strain relations:** Strain relations, Distinction between Elastic and Plastic Stress- Strain Relations, Plastic work, Derivation of Plastic Stress- strain relations. (6 Lectures)
- **Elastoplastic Problems of Spheres and Cylinders:** General relations, thick hollow sphere with internal pressure and thermal loading, Hollow sphere- Spread of Plastic Zone, Residual Stresses and Strain hardening material. (8 Lectures)
- **Plane problems in plasticity:** Beltrami- Michell equation, Plastic Bending of Plates, Deflection of Circular Plates, Plane Strain Analogy for Plate Bending, Yield Line theory for Plates, Axis symmetric case, General theorems of plasticity, Drucker's postulates, Integration of Plasticity Equation. (8 Lectures)
- **The Torsion problem:** Torsion of Prismatic Bar, General relations, elasticity solutions, perfect plasticity, elastoplastic torsion with strain hardening, bar with rectangular cross-section, bar with circular cross-section. (6 Lectures)
- **Module VII:** Slip-Line Field and limit analysis. (4 Lectures)

Text Books

1. J. Chakrabarty, **Applied Plasticity**, 2nd Edition, Springer.
2. Alexander Mendelson, **PLASTICITY: Theory and Application**, The Macmillan Company, New York.
3. Jacob Lubliner, **Plasticity Theory**, Courier Corporation, 2008.

References

1. R. M. Jones, **Deformation theory of plasticity**, Bull Ridge Publishing.
2. Akhtar S Khan Sujain Huang, **Continuum Theory of Plasticity**, John Wiley and sons.
3. L.M. Kachanov, **Fundamentals Theory of Plasticity**, Dover Publication.
4. Jacob Lubliner, **Plasticity theory**, Dover Publications.